

YONGGANG JIANG

GitHub: yonggangjiang Google Scholar: Yonggang Jiang

Max Planck Institute for Informatics (MPI-INF), E14 Room 315, Saarland Informatics Campus

E-mail: yjiang@mpi-inf.mpg.de Homepage: yonggangjiang.github.io

EDUCATION

Max Planck Institute for Informatics PhD in Computer Science Advisor: Danupon Nanongkai	<i>Apr. 2023 – Present</i>
Max Planck Institute for Informatics Preparatory Phase for Doctoral Study	<i>Oct. 2021 – Mar. 2023</i>
Nanjing University B.S. in Computer Science and Technology GPA: 94.4/100, Ranking: 1/31	<i>Sep. 2017 – Jul. 2021</i>
University of California, Berkeley Berkeley International Study Program GPA: 4.0/4.0	<i>Jan. 2020 – May 2020</i>

HONORS & AWARDS

Google PhD Fellowship	<i>2025</i>
Distinguished Paper Award, SPAA 2025	<i>2025</i>
Outstanding Graduate, Nanjing University	<i>2021</i>
National Elite Program Scholarship, Outstanding Prize (top 1)	<i>2018, 2019, 2020</i>
China Collegiate Programming Contest (CCPC), Gold Medal	<i>2018</i>
National Olympiad in Informatics in Provinces (NOIP), First Prize	<i>2015</i>

PUBLICATIONS

* Following the convention in mathematics, authors are listed in alphabetical order with equal contribution.

- [1] A Subquadratic Two-Party Communication Protocol for Minimum Cost Flow
Hossein Gholizadeh, Yonggang Jiang.
In submission.
- [2] Parallel Small Vertex Connectivity in Near Linear Work and Polylogarithmic Depth
Yonggang Jiang, Changki Yun.
In submission.
- [3] Perfect Simulation of Las Vegas Algorithms via Local Computation
Xinyu Fu, Yonggang Jiang, Yitong Yin.
In submission.
- [4] Reviving Thorup's Shortcut Conjecture
Aaron Bernstein, Bernhard Haeupler, Gary Hoppenworth, Yonggang Jiang, Maximilian Probst Gutenberg, Seth Pettie, Thatchaphol Saranurak, Leon Schiller.
In submission.
- [5] Reducing Shortcut and Hopset Constructions to Shallow Graphs
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
SIAM Symposium on Simplicity in Algorithms (SOSA) 2026.

- [6] Directed and Undirected Vertex Connectivity Problems are Equivalent for Dense Graphs
Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
SIAM Symposium on Simplicity in Algorithms (SOSA) 2026.
- [7] Minimum s-t Cuts with Fewer Cut Queries
Danupon Nanongkai, Yonggang Jiang, Pachara Sawettamalya.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.
- [8] Shortcuts and Transitive-Closure Spanners Approximation
Parinya Chalermsook, Yonggang Jiang, Sagnik Mukhopadhyay, Danupon Nanongkai.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.
- [9] New Oracles and Labeling Schemes for Vertex Cut Queries
Yonggang Jiang, Merav Parter, Asaf Petruschka.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.
- [10] Parallel $(1 + \epsilon)$ -Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work
Bernhard Haeupler, Yonggang Jiang, Yaowei Long, Thatchaphol Saranurak, Shengzhe Wang.
IEEE Symposium on Foundations of Computer Science (FOCS) 2025.
- [11] Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances
Jan van den Brand, Hossein Gholizadeh, Yonggang Jiang, Tijn de Vos.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2025.
Distinguished Paper Award
- [12] Deterministic Vertex Connectivity via Common-Neighborhood Clustering and Pseudorandomness
Chaitanya Nalam, Yonggang Jiang, Thatchaphol Saranurak, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.
- [13] Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations
Joakim Blikstad, Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.
- [14] Parallel and Distributed Exact Single-Source Shortest Paths with Negative Edge Weights
Vikrant Ashvinkumar, Aaron Bernstein, Nairen Cao, Christoph Grunau, Bernhard Haeupler, Yonggang Jiang, Danupon Nanongkai, Hsin Hao Su.
European Symposium on Algorithms (ESA) 2024.
- [15] Finding a Small Vertex Cut on Distributed Networks
Yonggang Jiang, Sagnik Mukhopadhyay.
ACM Symposium on Theory of Computing (STOC) 2023.
- [16] Robust and Optimal Contention Resolution without Collision Detection
Yonggang Jiang, Chaodong Zheng.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2022.
- [17] Tight Trade-off in Contention Resolution without Collision Detection
Haimin Chen, Yonggang Jiang, Chaodong Zheng.

RESEARCH VISITS & INVITED TALKS

National University of Singapore, Singapore, Hosted by Yi-Jun Chang. Talk: Parallel $(1 + \epsilon)$ -Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work	<i>Sep. 2025</i>
New York University, New York, NY, USA Hosted by Aaron Bernstein. Talk: Parallel $(1 + \epsilon)$ -Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work	<i>Sep. 2025</i>
Carnegie Mellon University, Pittsburgh, PA, USA Hosted by Jason Li. Talk: Parallel $(1 + \epsilon)$ -Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work	<i>Aug. 2025</i>
University of Michigan, Ann Arbor, MI, USA Hosted by Thatchaphol Saranurak.	<i>Jul. 2025</i>
ETH Zurich, Zurich, Switzerland Hosted by Sorrachai Yingchareonthawornchai.	<i>Feb. 2025</i>
Tsinghua University, Beijing, China Hosted by Ran Duan. Talk: New Tools and Faster Algorithms for Vertex Min-Cut	<i>Dec. 2024</i>
Nanjing University, Nanjing, China Hosted by Yitong Yin. Talk: New Tools and Faster Algorithms for Vertex Min-Cut	<i>Dec. 2024</i>
Shanghai University of Finance and Economics, Shanghai, China Hosted by Pinyan Lu. Talk: New Tools and Faster Algorithms for Vertex Min-Cut	<i>Dec. 2024</i>
ETH Zurich, Zurich, Switzerland Hosted by Weiming Feng. Talk: New Tools and Faster Algorithms for Vertex Min-Cut	<i>Nov. 2024</i>
INSAIT, Sofia, Bulgaria Hosted by Bernhard Haeupler.	<i>Sep. 2024</i>
University of Warwick, Coventry, UK Hosted by Sayan Bhattacharya. Talk: Deterministic k-Vertex Connectivity in k Max-Flows	<i>Jun. 2024</i>
University of Sheffield, Sheffield, UK Hosted by Sagnik Mukhopadhyay.	<i>Jun. 2024</i>
Nanjing University, Nanjing, China Hosted by Penghui Yao.	<i>Nov. 2023</i>
Rutgers University, New Brunswick, NJ, USA Hosted by Aaron Bernstein.	<i>Jun. 2023</i>
Aalto University, Helsinki, Finland Hosted by Parinya Chalermsook.	<i>Apr. 2023</i>

Aalto University, Helsinki, Finland
Hosted by Parinya Chalermsook.

Dec. 2022

University of Sheffield, Sheffield, UK
Hosted by Sagnik Mukhopadhyay.

May 2022

CONFERENCE & WORKSHOP TALKS

Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances

SPAA 2025, Portland, OR, USA

Jul. 2025

Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations

STOC 2025, Prague, Czech Republic

Jun. 2025

Contributed Talk: Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations

HALG 2025, Zurich, Switzerland

Jun. 2025

Contributed Talk: Deterministic k-Vertex Connectivity in k Max-Flows

HALG 2024, Warsaw, Poland

Jun. 2024

Parallel, Distributed, and Quantum Exact Single-Source Shortest Paths with Negative Edge Weights

ESA 2024, London, UK

Jun. 2024

Finding a Small Vertex Cut on Distributed Networks

STOC 2023, Orlando, FL, USA

Jun. 2023

Robust and Optimal Contention Resolution without Collision Detection

SPAA 2022, online

Jul. 2022

Tight Trade-off in Contention Resolution without Collision Detection

PODC 2021, online

Jul. 2021

TEACHING EXPERIENCE

Seminar: Foundations of Machine Learning, MPI-INF

Role: Coordinator and Speaker.

Summer 2023

Data Structures and Algorithms, Nanjing University

Role: Teaching Assistant.

Fall 2020, Fall 2019

STUDENTS & MENTORING

Ioannis Dorkofikis (M.Sc.)

Internship at MPI-INF, from NTUA, Greece.

Spring 2025

Changki Yun (B.Sc.)

Internship at MPI-INF, from SNU, Korea.

Outcome: Paper on parallel vertex connectivity (in submission)

Fall 2024

Hossein Gholizadeh (B.Sc.)

Bachelor's Thesis Internship at MPI-INF, from KIT, Germany.

Outcome: Paper in SPAA 2025 (Distinguished Paper Award)

Summer 2024

Alireza Danaei (B.Sc.)

Internship at MPI-INF, from Sharif Univ., Iran

Summer 2023

Vassilis Xanthopoulos (M.Sc.)

Master's Thesis Internship at MPI-INF, from NTUA, Greece

Spring 2023